

Chapter 4

Implementation and Result Analysis

4.1. Implementation Overview

The complete pipeline was implemented across 12 separate Google Colab notebooks to ensure modularity, ease of debugging, and clear separation of concerns while maintaining structural similarity for fair comparison across the 32 clustering combinations. The notebooks were organised as follows:

Notebooks 1–4: Data Preprocessing and Encoding These notebooks handled initial data preparation, conditional API-based filling, and multi-strategy encoding. Notebook 1 performed data cleaning (removing duplicates, standardising categories, generating missing value reports, dropping unnecessary columns such as source URLs and full names). Notebook 2 conditionally filled latitude and longitude using hometown information via API calls. Notebook 3 conditionally filled weather-related columns (e.g., `temp_min`, `temp_max`, `feels_like`, `air_humidity`, `wind_deg`, `clouds_sky`) using latitude, longitude, unix time, and date via API. Notebook 4 applied multiple encoding strategies tailored to different column types (e.g., one-hot, frequency, label, ordinal).

Notebooks 5–7: Imputation One notebook each for MICE, KNN, and MissForest, handling both context-aware and no-context predictor strategies to produce the 6 imputed datasets (1613 rows $\times$ 37 columns each).

Notebooks 8–11: Clustering One notebook each for K-Means, Agglomerative Clustering, DBSCAN, and Spectral Clustering.

Notebook 12: Evaluation, Ranking, and Cluster Profiling Comprehensive evaluation, composite scoring, ranking, group-level comparisons, metric visualisations, and detailed cluster profiling.

The imputation notebooks prepared the 6 imputed datasets, while clustering notebooks loaded the 8 final datasets (6 imputed + 2 non-imputed baselines: `df_encoded_20` with 853 rows/22 columns and `df_encoded_40` with 403 rows/30 columns). Preprocessing applied `StandardScaler` for numerical features and optional PCA (retaining significant variance, e.g., 99% or fixed `n_components` like 10/20). Hyperparameter tuning used grid search and fine-tuning, with evaluation via the multi-metric framework from Section 3.6. Optimal configurations were selected, and cluster labels appended to the original DataFrames (e.g., `'kmeans_cluster'`, `'agg_cluster'`, `'dbscan_cluster'`, `'spc_cluster'`).

Batch processing was enabled by organising datasets into logical groups (e.g., `agg_dfs`, `dbscan_dfs`, `kmeans_dfs`, `spc_dfs` for algorithm-specific analysis; `context_dfs`, `no_context_dfs`, `no_imputed_dfs`, `mice_dfs`, `rf_dfs`, `knn_dfs` for cross-imputation comparisons), allowing systematic evaluation across strategies, imputation methods, and algorithm types. Key libraries included `scikit-learn` (for preprocessing, clustering, and metrics), `NumPy`, `SciPy` (`cdist` for Dunn/Xie-Beni), `pandas`, and `matplotlib/seaborn` for visualisation. All experiments used a fixed `random_state=42` to ensure full reproducibility.

This modular approach facilitated incremental development, isolated testing of each component, and consistent application of the pipeline, minimising implementation bias across the 32 combinations.

4.2. Cluster Quality Comparison

4.2.1. Initial Manual Evaluation

Initial evaluation of the 32 clustering outputs revealed significant variation in quality. For example, some DBSCAN configurations produced high silhouette scores (e.g., 0.8744) but formed only tiny clusters (e.g., two clusters of 3 points each, with 1611/1617 points labelled as noise). This highlighted a common issue: geometric metrics can favour small, compact clusters while ignoring excessive noise and poor data utilisation. Similar patterns appeared across algorithms, where high scores on a few points masked impractical results (e.g., one dominant cluster with many singletons). These observations underscored the need for the proposed balance metric and composite scoring to penalise such cases and prioritise interpretable, well-utilised clusters.

4.2.2. Composite Scoring and Ranking

The 32 clustering combinations were ranked using the composite score framework described in Section 3.6. Figure 4.1 presents a heatmap of the normalised metric values (min-max scaling per metric) and composite scores, sorted by final rank. Higher values appear redder, lower values bluer. Detailed raw (non-normalised) metric values for all cases are provided in Supplementary Table S1.

The ranking reveals clear patterns in performance. Imputed datasets dominate the top positions, with all top 10 cases being imputed (6 from no-context strategies and 4 from context-aware). This indicates that imputation substantially improves clustering quality compared to baselines. Among imputed cases, no-context variants (using all columns as predictors) show a slight advantage over context-aware ones (using only correlated predictors), as they hold 6 of the top 10 spots. This suggests that maximising available information during imputation provides a modest but consistent benefit.

Compare to MICE and KNN MissForest imputation is working far better since 6 of the top 10 cases are imputed with MissForest with 2 from MICE and 2 KNN. So we can see that the MissForest imputation technique is working better than other two methods.

Algorithm diversity is evident in the top 10, with representatives from all four clustering methods. Spectral Clustering (labelled ‘spc’) leads with 5 cases in the top 10, followed by Agglomerative and K-Means. The single best-performing combination is Agglomerative Clustering with MissForest no-context imputation (agg_rf_no_context, rank 1, composite score 27.30), demonstrating excellent separation and balance — the maximum achievable composite score is 32, so 27.30 sits close to the top of the scale. High-noise or severely imbalanced cases (e.g., DBSCAN with tiny clusters) consistently ranked at the bottom, underscoring the balance metric’s critical role in penalising impractical configurations. The worst-performing pipeline was DBSCAN with KNN imputation under the context-aware predictor strategy, scoring 4.20 — very close to the theoretical minimum of 1 on the 1–32 scale.

These patterns are analysed in more depth in the group-level comparisons (Section 4.4) and the discussion of findings (Section 4.7).

(Rows sorted by final rank; columns: balance, silhouette, Calinski-Harabasz, Dunn, Davies-Bouldin (inverted), Xie-Beni (inverted), composite score. Green = higher values, red = lower values.)

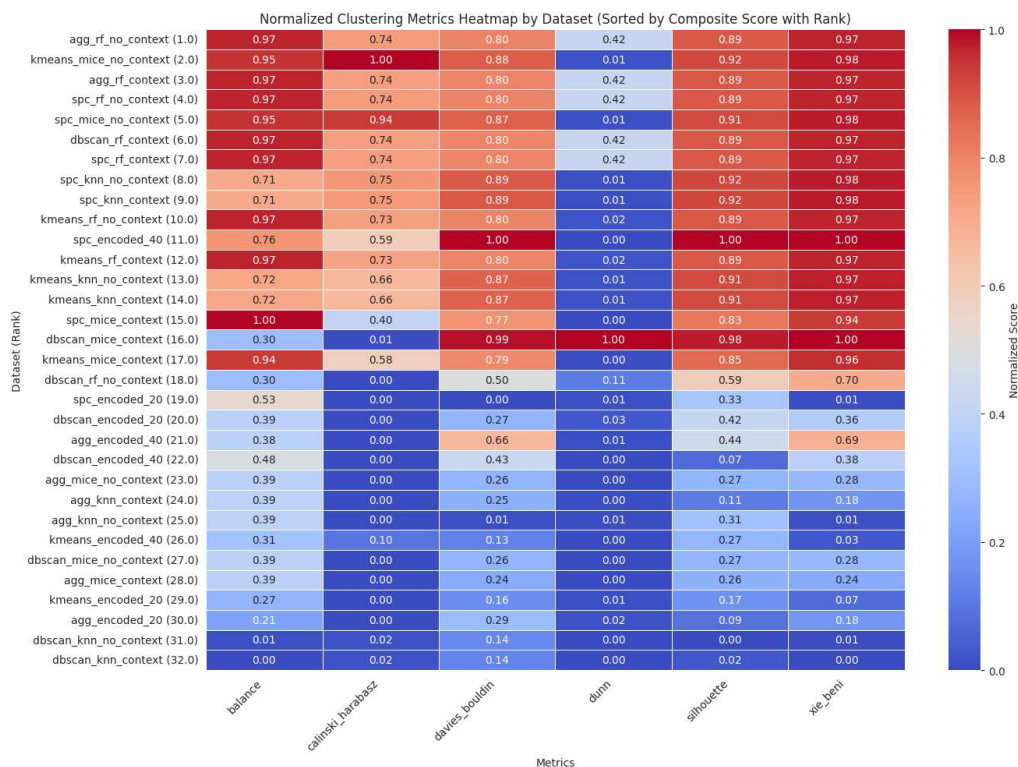


Figure 4.1: Composite Scoring and Ranking

4.2.3. Visual Comparison of Combined Scores

Figure 4.2 presents a horizontal bar graph of the composite scores for all 32 clustering combinations, sorted by final rank (highest to lowest). Imputed pipelines consistently outperformed the non-imputed baselines. This visualisation underscores the substantial improvement in overall clustering quality achieved through imputation.

Figure 4.3 displays a grid of PCA-reduced scatter plots for the top 16 ranked cases, arranged in 4 rows and 4 columns (sorted by descending composite rank from top-left to bottom-right). Each subplot shows the data points coloured by their assigned cluster labels after dimensionality reduction to the first two principal components. The top-ranked cases (upper rows) exhibit clear, well-separated clusters with minimal overlap, indicating strong structure and high-quality partitioning. In contrast, the lower-ranked cases within this selection (lower rows) display increasingly messy, heavily overlapping clusters, often with signs of fragmentation or poor separation. Full scatter plots for all 32 cases are provided in Supplementary Figure S2.

Figure 4.4 shows a bar graph of the number of unique clusters (excluding noise labels of -1 in DBSCAN) for the top 16 ranked cases. Top-performing cases predominantly form 2 balanced clusters, reflecting interpretable and well-distributed partitioning. Lower-ranked cases within this selection tend to produce either very few clusters (due to excessive noise) or excessively many (over-segmentation), both of which correlate with reduced composite scores. Full cluster count data for all 32 cases are provided in Supplementary Figure S3.

These visual comparisons highlight that the proposed composite scoring framework effectively prioritises configurations with meaningful, evenly utilised clusters while penalising those with geometric artifacts or impractical distributions.

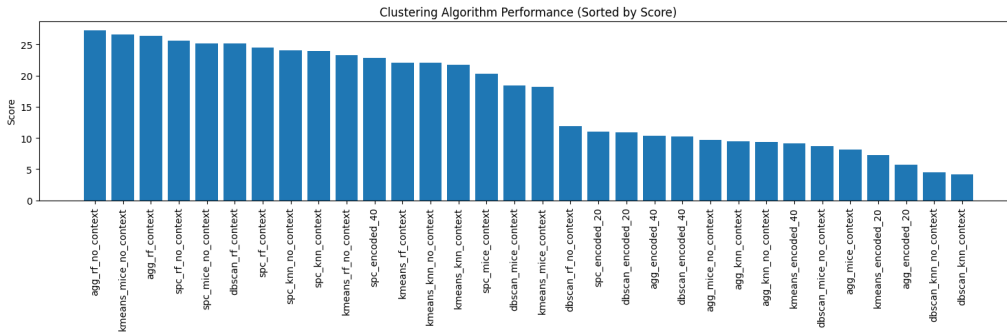


Figure 4.2: Horizontal bar graph of composite scores for all 32 clustering combinations, sorted by final rank (highest to lowest).

4.3. Optimal Parameters and Performance Summary

Table 4.1 summarises the optimal hyperparameters selected via grid search and fine-tuning for each of the 32 cases, along with key performance metrics and ranking. Across

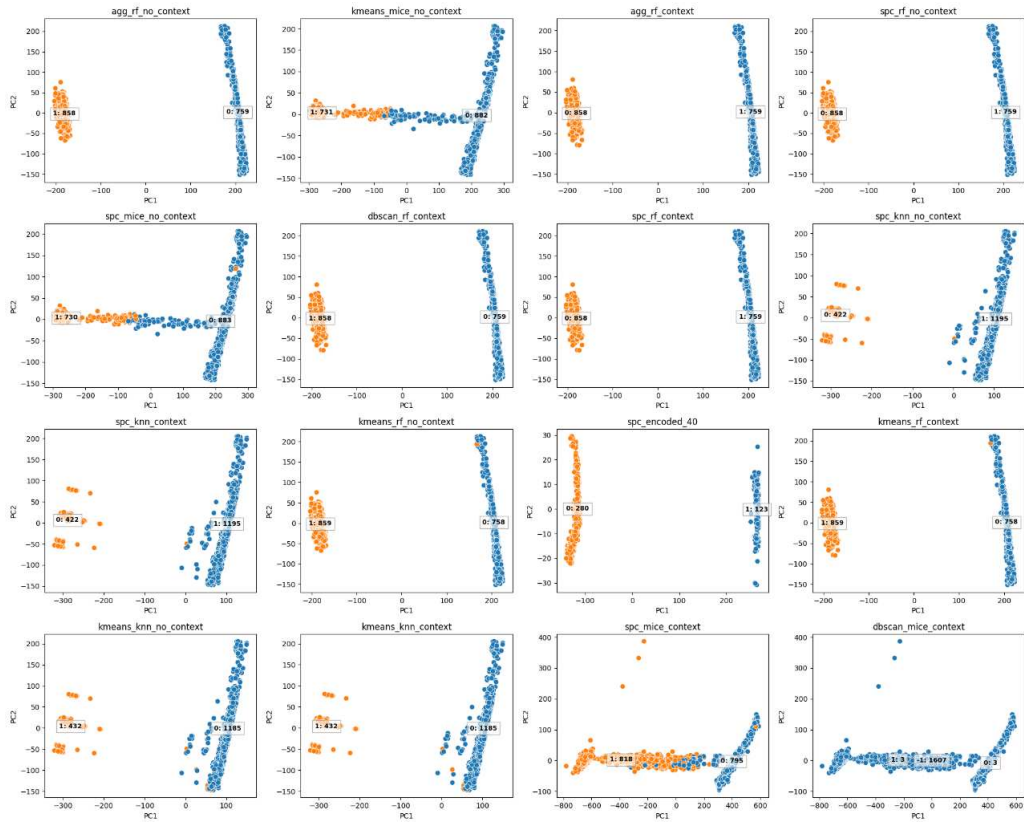


Figure 4.3: Grid of PCA-reduced scatter plots (4 rows $\times$ 4 columns) for the top 16 ranked cases, sorted by descending composite rank (top-left = best). Points are coloured by cluster labels.

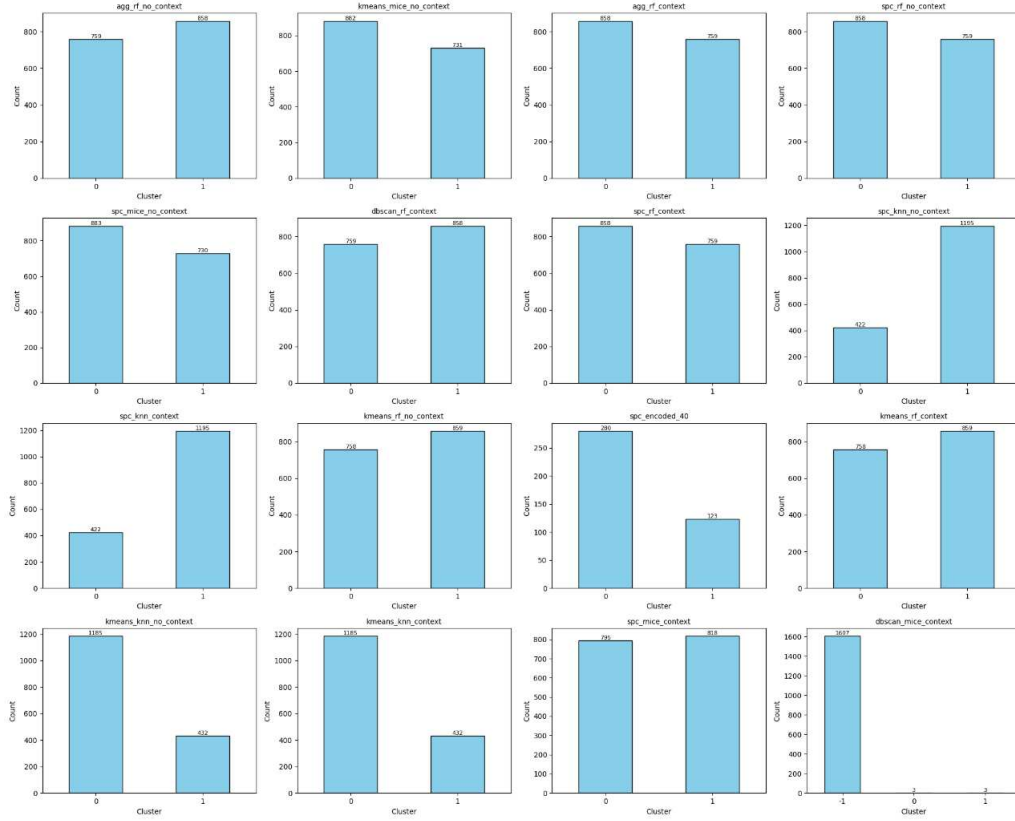


Figure 4.4: Horizontal bar graph showing the number of unique clusters (excluding noise in DBSCAN) for the top 16 ranked cases.

nearly all datasets, the best-performing solutions consistently favoured two clusters ($n_clusters = 2$) for K-Means, Agglomerative, and Spectral clustering, indicating an inherent binary structure in the data regardless of imputation or context usage. Spectral clustering showed complete parameter stability, always selecting $affinity = nearest_neighbors$ with $assign_labels = kmeans$, while K-Means relied on low $n_clusters$ and moderate n_init , except for encoded-only datasets where higher cluster counts were needed. Context-aware imputation did not substantially change optimal hyperparameters, suggesting it mainly improved data quality rather than cluster structure. In contrast, DBSCAN was highly sensitive to data representation, with cosine distance dominating for encoded and RF/MICE datasets, and larger eps values required for KNN-imputed data.

Rank	Abbreviation	Optimal Parameters	Silhouette	Composite
1	agg_rf_no_context	n_clusters=2, linkage=complete, metric=cosine	0.756380	27.30
2	kmeans_mice_no_context	n_clusters=2, init=k-means++, n_init=10, max_iter=300	0.792085	26.60
3	agg_rf_context	n_clusters=2, linkage=average, metric=cosine	0.754761	26.35
4	spc_rf_no_context	n_clusters=2, affinity=nearest_neighbors, assign_labels=kmeans	0.756380	25.60
5	spc_mice_no_context	n_clusters=2, affinity=nearest_neighbors, assign_labels=kmeans	0.787801	25.20
6	dbscan_rf_context	eps=0.3, min_samples=3, metric=cosine	0.754761	25.15
7	spc_rf_context	n_clusters=2, affinity=nearest_neighbors, assign_labels=kmeans	0.754761	24.45
8	spc_knn_no_context	n_clusters=2, affinity=nearest_neighbors, assign_labels=kmeans	0.795369	24.10
9	spc_knn_context	n_clusters=2, affinity=nearest_neighbors, assign_labels=kmeans	0.794904	24.00
10	kmeans_rf_no_context	n_clusters=2, init=k-means++, n_init=10, max_iter=300	0.755367	23.30
11	spc_encoded_40	n_clusters=2, affinity=nearest_neighbors, assign_labels=kmeans	0.903649	22.80
12	kmeans_rf_context	n_clusters=2, init=k-means++, n_init=10, max_iter=300	0.753743	22.10
13	kmeans_knn_no_context	n_clusters=2, init=k-means++, n_init=10, max_iter=300	0.785370	22.05
14	kmeans_knn_context	n_clusters=2, init=k-means++, n_init=10, max_iter=300	0.784890	21.75
15	spc_mice_context	n_clusters=2, affinity=nearest_neighbors, assign_labels=kmeans	0.670781	20.30
16	dbscan_mice_context	eps=0.3, min_samples=3, metric=cosine	0.874364	18.45
17	kmeans_mice_context	n_clusters=2, init=random, n_init=20, max_iter=300	0.708758	18.25
18	dbscan_rf_no_context	eps=0.3, min_samples=15, metric=cosine	0.351424	11.90
19	spc_encoded_20	n_clusters=2, affinity=nearest_neighbors, assign_labels=kmeans	0.012104	11.00
20	dbscan_encoded_20	eps=0.5, min_samples=3, metric=cosine	0.121859	10.85
21	agg_encoded_40	n_clusters=2, linkage=complete, metric=manhattan	0.157153	10.35
22	dbscan_encoded_40	eps=0.3, min_samples=10, metric=cosine	-0.339146	10.20
23	agg_mice_no_context	n_clusters=2, linkage=complete, metric=euclidean	-0.077905	9.65
24	agg_knn_context	n_clusters=2, linkage=complete, metric=manhattan	-0.285265	9.45
25	agg_knn_no_context	n_clusters=2, linkage=complete, metric=manhattan	-0.019878	9.35
26	kmeans_encoded_40	n_clusters=8, init=k-means++, n_init=50, max_iter=300	-0.071859	9.15
27	dbscan_mice_no_context	eps=0.3, min_samples=3, metric=cosine	-0.077905	8.65
28	agg_mice_context	n_clusters=2, linkage=complete, metric=manhattan	-0.094494	8.15
29	kmeans_encoded_20	n_clusters=10, init=k-means++, n_init=50, max_iter=300	-0.212661	7.20
30	agg_encoded_20	n_clusters=6, linkage=average, metric=euclidean	-0.316824	5.65
31	dbscan_knn_no_context	eps=1.9, min_samples=3, metric=euclidean	-0.436810	4.50
32	dbscan_knn_context	eps=1.9, min_samples=3, metric=euclidean	-0.405025	4.20

Table 4.1: Optimal Parameters and Performance for 32 Cases

4.4. Group-Level Analysis

4.4.1. Metric-Specific Performance

To better understand the drivers behind the composite ranking, this section analyses the performance of all 32 cases under each individual clustering quality metric. All metrics were normalised using min–max scaling across the full set of cases to ensure comparability. For each metric, we examine systematic patterns across imputation strategies, context usage, and clustering algorithms, and discuss how these patterns contribute to the final composite score.

Balance:

The balance metric shows a clear separation between imputed and non-imputed (baseline) cases. Most imputed pipelines—particularly those using MissForest and MICE—achieve

consistently high balance scores, indicating well-distributed cluster sizes. In contrast, DBSCAN-based pipelines and encoded-only baselines tend to score poorly, reflecting highly imbalanced or fragmented cluster assignments. Since balance is heavily penalised for extreme cluster size disparities, low values in this metric significantly depress the composite scores of DBSCAN configurations, even when other metrics perform moderately well. Overall, strong balance performance is a necessary condition for achieving a high composite rank.

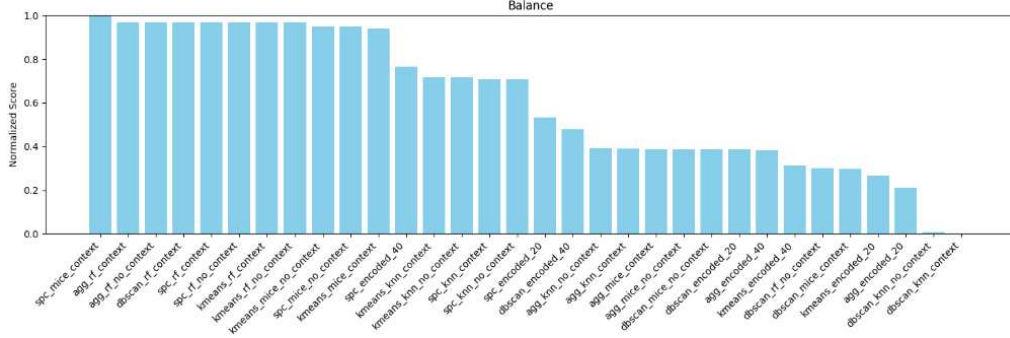


Figure 4.5: Normalised Balance scores across all 32 clustering pipelines.

Silhouette Score

Silhouette scores exhibit strong discrimination among the 32 cases and play a major role in ranking separation. Spectral Clustering and K-Means with imputation—especially no-context variants—dominate the upper range, indicating compact and well-separated clusters. Context-aware imputation performs competitively but does not consistently outperform no-context variants. Encoded baselines and DBSCAN pipelines frequently produce near-zero or negative silhouette values, signaling overlapping or poorly defined clusters. As a core measure of intra- vs. inter-cluster separation, silhouette score strongly reinforces the dominance of imputed pipelines in the composite ranking.

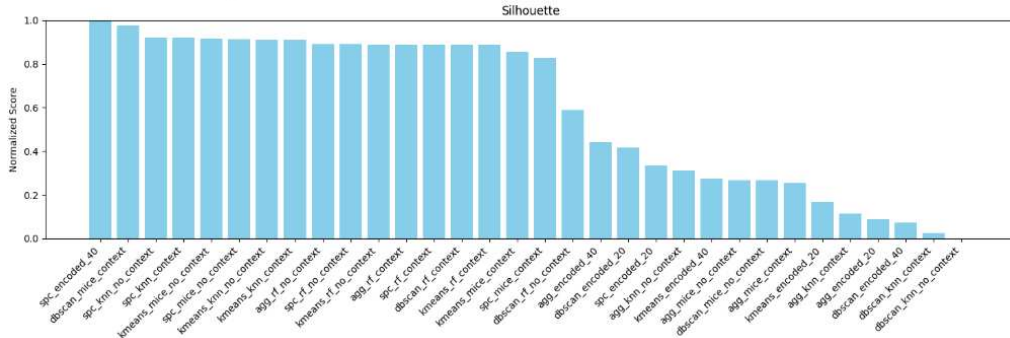


Figure 4.6: Normalised Silhouette scores across all 32 clustering pipelines.

The Calinski–Harabasz index sharply distinguishes high-quality global cluster structure from weak or degenerate solutions. Imputed K-Means, Spectral, and Agglomerative pipelines achieve values several orders of magnitude higher than encoded and DBSCAN-based cases. This metric particularly favours centroid-based and hierarchical methods operating on well-imputed feature spaces. Extremely low Calinski–Harabasz values among DBSCAN and encoded cases contribute heavily to their low composite scores, making this metric a major driver of the top–bottom separation observed in the overall ranking.



The Dunn index displays a highly skewed distribution, with only a small subset of cases achieving meaningful values. Agglomerative and Spectral pipelines with MissForest imputation stand out, producing exceptionally high Dunn scores that indicate strong minimum inter-cluster separation relative to cluster diameter. Most other pipelines—including K-Means and all encoded baselines—yield values close to zero, reflecting limited separation guarantees. Because of its sensitivity to minimum-distance relationships, the Dunn index strongly rewards structurally clean cluster solutions and disproportionately boosts the top-ranked imputed cases in the composite score.



Davies–Bouldin Index

After normalisation (and inversion for comparability), the Davies–Bouldin index shows patterns broadly consistent with silhouette and Dunn metrics. Imputed Spectral and K-Means pipelines generally achieve favourable scores, indicating low average cluster similarity. Encoded and DBSCAN-based pipelines perform poorly, reflecting high overlap between clusters. While Davies–Bouldin alone does not fully reorder the rankings, it reinforces penalties for noisy or overlapping cluster structures and contributes to the demotion of DBSCAN and baseline cases in the composite evaluation.

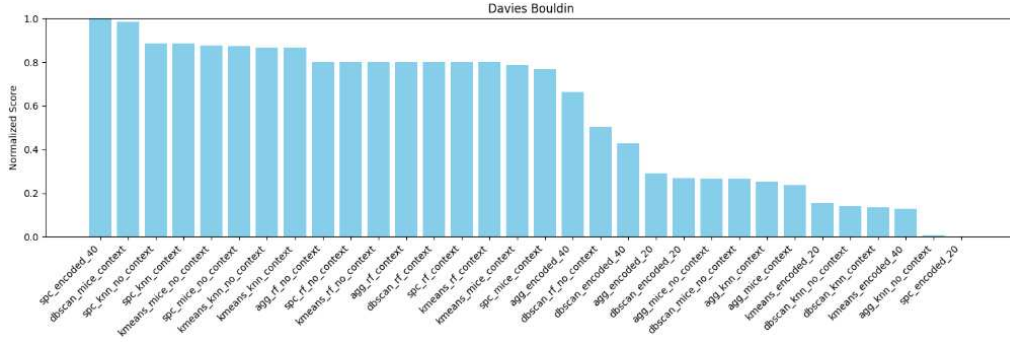


Figure 4.9: Normalised Davies–Bouldin index across all 32 clustering pipelines.

Xie–Beni Index

The Xie–Beni index strongly favours imputed pipelines, particularly those using MissForest and MICE, across both context-aware and no-context settings. High-performing cases exhibit low intra-cluster variance relative to inter-cluster separation, while encoded and DBSCAN configurations score near zero after normalisation. This metric is especially effective at identifying degenerate clustering outcomes, such as tiny or highly scattered clusters, and therefore plays a critical role in suppressing artificially optimistic scores from unstable pipelines. Its behaviour closely aligns with the final composite ranking.

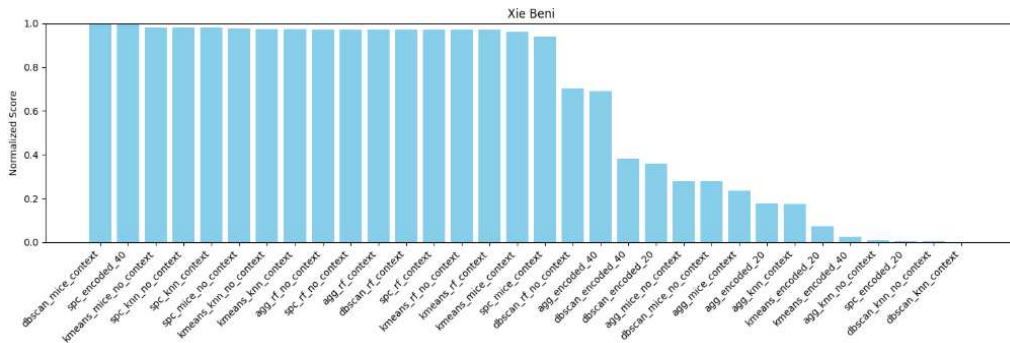


Figure 4.10: Normalised Xie–Beni index across all 32 clustering pipelines.

4.4.2. Group Average Comparisons

To complement the case-level analysis, we further examined group-level average performance by aggregating metrics across predefined groups. These comparisons highlight systematic effects of imputation strategy, context-aware feature selection, and clustering algorithm choice. For each comparison, average values of all six normalised metrics were computed, and corresponding bar plots are presented to support visual interpretation.

4.4.2.1. Comparison of Imputation Strategies

MissForest imputation substantially outperforms all other strategies, achieving the highest average scores in balance (0.64), silhouette (0.70), Calinski–Harabasz (8481), Dunn (0.66), and Xie–Beni (0.93), as well as the highest average composite score (23.27). This indicates that RF-based imputation preserves cluster structure while reducing noise and improving compactness. MICE shows moderate improvements over non-imputed baselines, particularly in balance and Dunn index, but remains clearly inferior to MissForest. KNN imputation performs inconsistently, with low Dunn (0.02) and weaker silhouette values, reflecting sensitivity to local noise. Non-imputed datasets consistently rank worst across all metrics, confirming that imputation is essential for stable clustering in this setting.

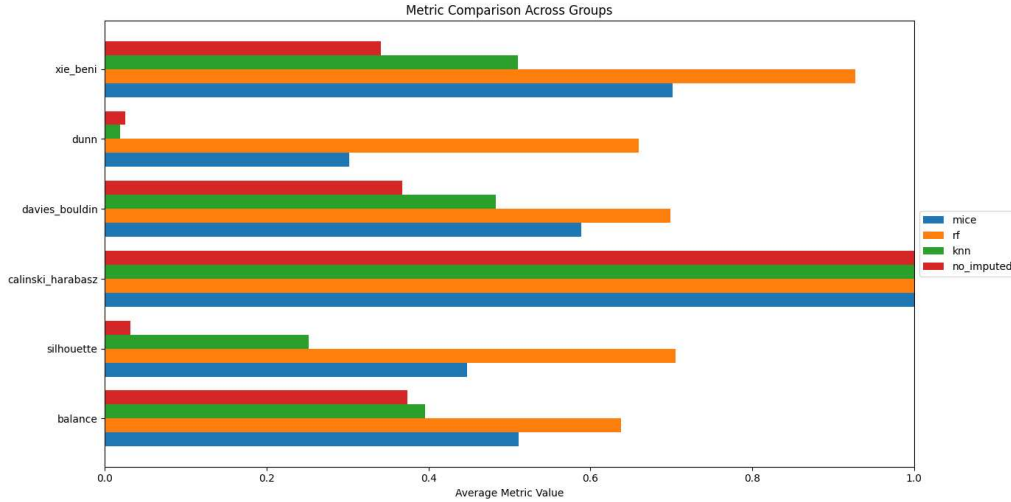


Figure 4.11: Average metric comparison across imputation strategies.

4.4.2.2. Context-Aware vs No-Context Imputation

Context-aware imputation yields higher averages for silhouette (0.51 vs 0.43), Dunn (0.45 vs 0.20), and Xie–Beni (0.76 vs 0.67), suggesting improved cluster separation and compactness when irrelevant predictors are excluded. In contrast, no-context imputation achieves a slightly higher Calinski–Harabasz score (6137 vs 5869), indicating that additional features may inflate variance-based separation even if they introduce noise. Balance

scores are comparable across both groups. Overall, context-aware imputation attains a marginally higher average composite score (18.55 vs 18.18), indicating a small but consistent advantage.

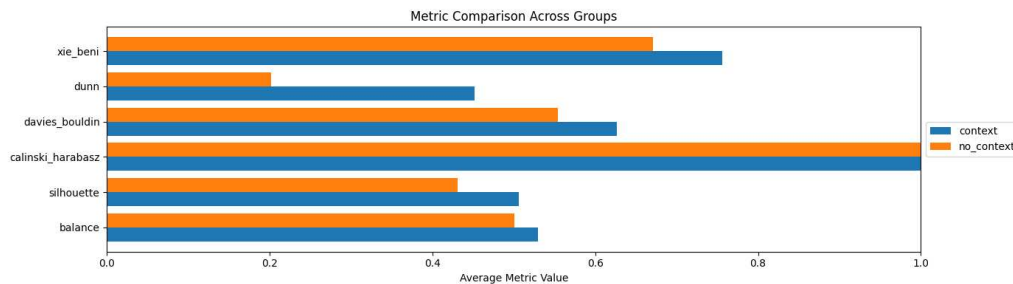


Figure 4.12: Average metric comparison between context-aware and no-context imputation.

4.4.2.3. Comparison of Clustering Algorithms

Spectral clustering achieves the highest average performance across most metrics, including silhouette (0.68), Calinski–Harabasz (8084), balance (0.60), and Xie–Beni (0.85), leading to the highest average composite score (22.18). K-Means performs competitively, particularly in balance (0.55) and Calinski–Harabasz (7346), but suffers from extremely low Dunn values (0.03), indicating weak minimum inter-cluster separation. Agglomerative clustering shows moderate balance and Dunn performance but lower silhouette scores overall. DBSCAN exhibits the weakest average performance, with low balance and silhouette scores despite a relatively higher Dunn index, reflecting its sensitivity to parameter settings and difficulty handling varying densities.

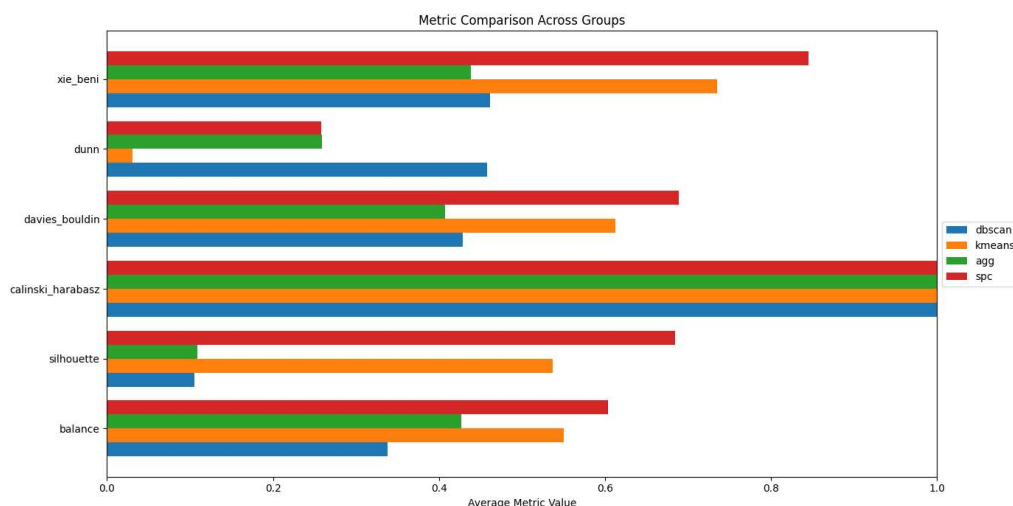


Figure 4.13: Average metric comparison across clustering algorithms.

4.5. Key Takeaways

This section highlights the main findings from the comparative evaluation of all 32 pipelines across imputation strategies, context-awareness settings, clustering algorithms, and clustering quality metrics.

Across all comparisons, imputation consistently improves clustering quality. Imputed pipelines substantially outperform non-imputed baselines, which achieve the lowest average composite score (10.90). Among imputation techniques, MissForest (RF) is the strongest performer, achieving the highest average composite score (23.27), followed by MICE (16.91), while KNN imputation trails behind (14.92). This confirms that tree-based imputation produces more stable and informative feature representations for clustering.

In terms of context awareness, context-aware pipelines achieve a slightly higher average composite score (18.55) than no-context pipelines (18.18). However, the difference is modest, and several of the top-ranked pipelines remain non-context-aware, indicating that using the full feature set during imputation can still be advantageous when strong imputation methods are applied.

When comparing clustering algorithms, Spectral Clustering (22.18) and K-Means (18.80) show higher average composite scores than Agglomerative (13.28) and DBSCAN (11.74). Despite Agglomerative Clustering achieving the single highest-ranked pipeline, its lower group average reflects greater sensitivity to preprocessing choices. DBSCAN shows the weakest overall performance, consistent with its instability under imbalance and noise.

From a metric perspective, balance and silhouette exert the strongest influence on composite rankings, effectively penalising pipelines that generate imbalanced or poorly separated clusters, even when other metrics appear favourable. This explains why pipelines with high individual metric scores (e.g., Dunn or Davies–Bouldin) may still rank poorly overall.

Finally, Table 4.2 summarises the best-performing pipeline for each comparison group, including clustering algorithm, imputation strategy, and context-awareness setting. This table provides a compact reference for the optimal configurations identified in this study and serves as a transition to the subsequent Findings and Discussion section.

Group Type	Group Name	Best Pipeline	Composite Score
Algorithm	Agglomerative	agg_rf_no_context	27.30
Algorithm	K-Means	kmeans_mice_no_context	26.60
Algorithm	DBSCAN	dbscan_rf_context	25.15
Algorithm	Spectral	spc_rf_no_context	25.60
Imputation	MissForest (RF)	agg_rf_no_context	27.30
Imputation	MICE	kmeans_mice_no_context	26.60
Imputation	KNN	spc_knn_no_context	24.10
Imputation	No Imputation	spc_encoded_40	22.80
Context	Context-aware	agg_rf_context	26.35
Context	No-context	agg_rf_no_context	27.30

Table 4.2: Best-performing pipeline for each comparison group.

4.6. Detailed Profiling of the Top-Ranked Cluster

From the comparative analysis presented earlier, the best-performing pipeline was identified as `agg_rf_no_context`, which combines Agglomerative Clustering with MissForest imputation using all available predictors (no contextual feature restriction). Given its superior composite score, this pipeline was selected for deeper inspection to determine whether the resulting clusters represent meaningful and interpretable structure within the data.

The `agg_rf_no_context` pipeline produced two distinct clusters: Cluster 0 containing 759 records and Cluster 1 containing 858 records, with no overlap and no noise points. This complete partitioning accounts for all 1,617 records in the dataset (Figures 4.3 and 4.4). The pipeline achieved a composite score of 27.30, supported by a balance score of 0.685 and a silhouette score of 0.756, indicating both strong separation and well-balanced cluster sizes. These characteristics confirm that the clustering outcome is well-suited for detailed interpretation. The following subsections profile the clusters across numerical, categorical, geographic, and temporal dimensions.

4.6.1. Decoding the Dataset

To enable interpretability, all label-encoded categorical features were decoded using the JSON mappings saved during preprocessing. This decoding step restores the original semantic values of each feature, allowing direct comparison of cluster-specific characteristics in their original context. The decoded dataset provides the foundation for analysing

differences between clusters across demographic, environmental, social, and geographic attributes.

4.6.2. Numerical Feature Profiling

Table 4.3 presents summary statistics (mean, median, and count) for all numerical features across the two clusters. To highlight the most informative patterns, Figures 4.14 and 4.15 present representative distribution plots for selected numerical variables that exhibited the strongest separation and domain relevance.

Feature	Cluster 0 (n=759)	Cluster 1 (n=858)
Age (mean / median)	28.00 / 24.00	25.80 / 23.00
Feels like (mean / median)	299.89 / 300.65	290.46 / 290.44
Temp min (mean / median)	298.95 / 299.51	23.78 / 25.61
Temp max (mean / median)	298.95 / 299.51	31.80 / 32.54
Air pressure (mean / median)	1004.77 / 1004.00	1007.01 / 1005.90
Humidity (mean / median)	85.19 / 87.00	71.71 / 70.11
Wind speed (mean / median)	3.52 / 3.00	7.27 / 7.20
Wind direction (mean / median)	149.05 / 145.00	193.32 / 194.53
Cloud cover (mean / median)	64.05 / 75.00	60.68 / 67.70
Latitude (mean / median)	23.89 / 23.92	23.67 / 23.76
Longitude (mean / median)	90.11 / 90.01	90.26 / 90.36
Day (mean / median)	16.78 / 16.00	14.72 / 15.00
Month (mean / median)	7.22 / 7.00	5.69 / 6.00
Year (mean / median)	2020 / 2020	2021.96 / 2022
Week (mean / median)	30.12 / 30.00	22.76 / 23.00

Table 4.3: Summary statistics (mean, median) for all numerical features across the two clusters. Temperature columns inherit raw source units: `feels_like`, `temp_min` and `temp_max` appear in Kelvin for Cluster 0 (which is dominated by the original 2020 Kaggle subset) and in Celsius for Cluster 1 (which is dominated by the manually collected 2017–2025 cases); identical `temp_min` and `temp_max` values in Cluster 0 reflect a source-data artefact of the Kaggle subset rather than a computation error.

Figure 4.14 illustrates the age distribution by cluster, revealing a clear demographic contrast, with one cluster consistently showing higher average age values. Figure 4.15 compares a temperature-related variable, `feels_like`, indicating distinct environmental conditions associated with each cluster. Secondary environmental features such as air humidity and wind speed also show consistent but less pronounced separation. These patterns suggest that the clustering captures coherent demographic–environmental structure rather than noise-driven variation.

Complete numerical feature comparisons for all variables are provided in Figure S4.

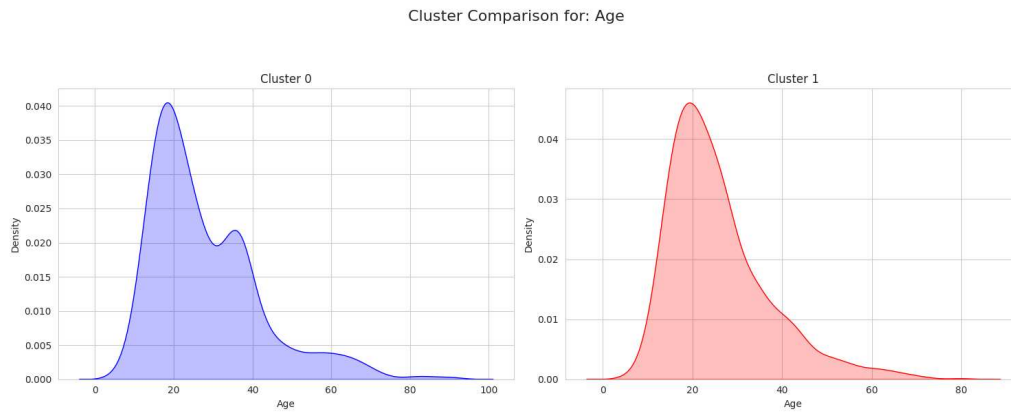


Figure 4.14: Comparing age for two clusters

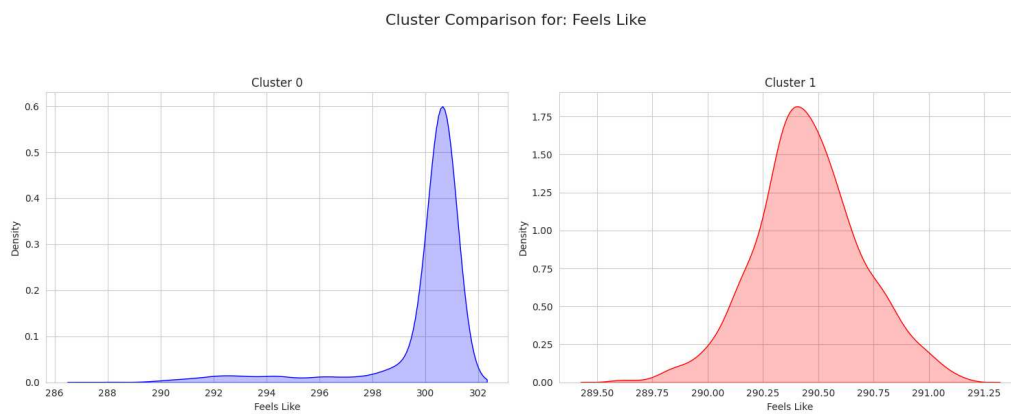


Figure 4.15: Comparing feels_like for two clusters

4.6.3. Categorical Feature Profiling

Categorical feature distributions were analysed to uncover social and contextual differences between clusters. Table 4.4 reports the most frequent categories for each categorical feature across clusters, while Figures 4.16 and 4.17 focus on a subset of the most discriminative variables to maintain interpretability.

Feature	Cluster 0 (Mode, %)	Cluster 1 (Mode, %)
Age group	Teen (48.5%)	Teen (39.2%)
Gender	Female (54.8%)	Male (50.8%)
Marital status	Married (48.4%)	Married (47.6%)
Profession group	Student (37.0%)	Student (39.7%)
Workplace	Semi-urban (54.5%)	Rural (50.0%)
Religion	Muslim (87.5%)	Muslim (85.4%)
Financial condition	Lower (54.3%)	Lower (57.6%)
Hometown	Bogura (4.0%)	Dhaka (9.1%)
Primary reason	Family issue (23.1%)	Mental (26.5%)
Method	Hanging (73.3%)	Hanging (74.1%)
Weather (main)	Clouds (38.1%)	Partly cloudy (99.9%)
Weather description	Haze (19.6%)	Haze (64.7%)
Weekday	Saturday (17.9%)	Sunday (16.9%)
Time of day	Night (36.9%)	Night (35.5%)

Table 4.4: Most frequent category (mode and percentage) for each categorical feature across the two clusters. The near-deterministic value for Cluster 1’s **weather_main** (Partly cloudy, 99.9%) reflects the dominant imputed value for the manually collected 2017–2025 records under MissForest rather than a homogeneous physical weather pattern.

Figure 4.17 compares workplace type for the two clusters, highlighting variation in social and environmental context. Figure 4.16 presents the distribution of reported reasons for suicide, revealing distinct dominant categories across clusters. Differences are also observed in weather related features like weather main (e.g., clear, clouds, rain) and weather description (e.g., broken clouds, clear sky, haze), indicating variation in environmental context.

Full categorical distributions across clusters are shown in Figure S5.

4.6.4. Geographic Profiling

Geographic distributions of the two clusters are visualised in Figure 4.18 using latitude–longitude heatmaps over Bangladesh. A single combined map with distinct colours for each cluster is used to emphasise spatial coherence and contrast.

The visualisation shows that the clusters are not spatially arbitrary. One cluster exhibits stronger concentration around major urban and densely populated regions (Cluster

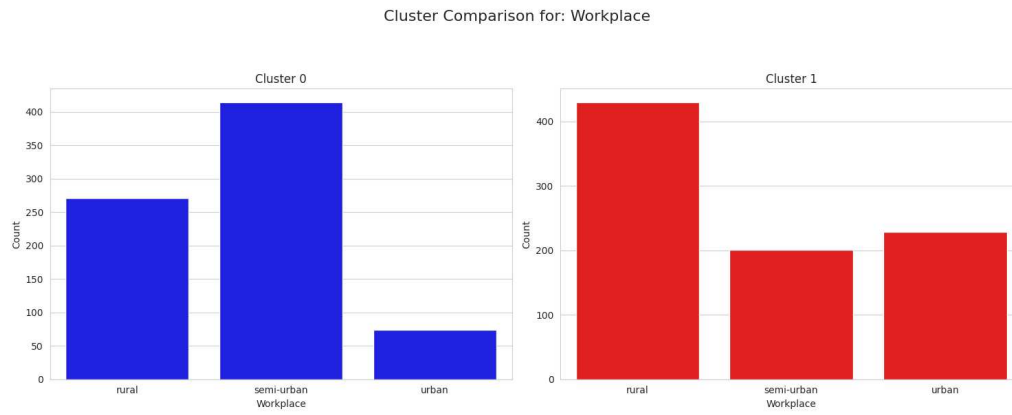


Figure 4.16: Comparing reason for two clusters

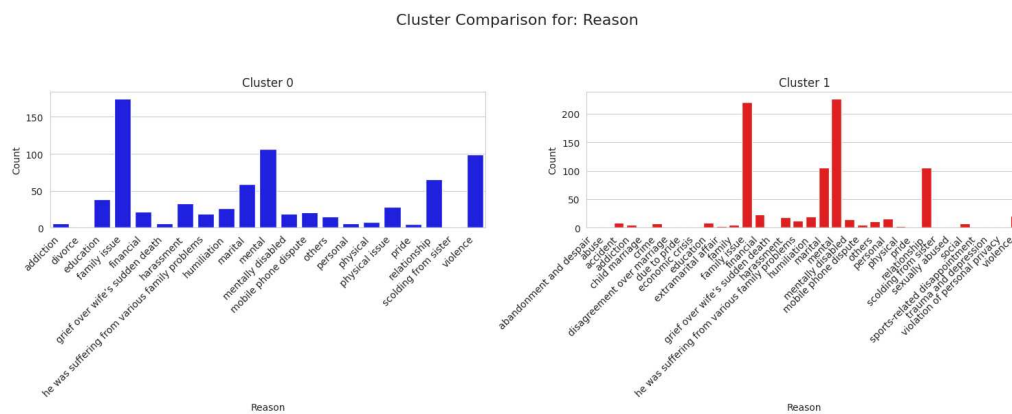


Figure 4.17: Comparing workplace for two clusters

1), while the other displays a more dispersed geographic pattern (Cluster 0). This spatial differentiation reinforces the numerical and categorical findings and suggests that geographic context contributes meaningfully to the cluster structure identified by the pipeline.

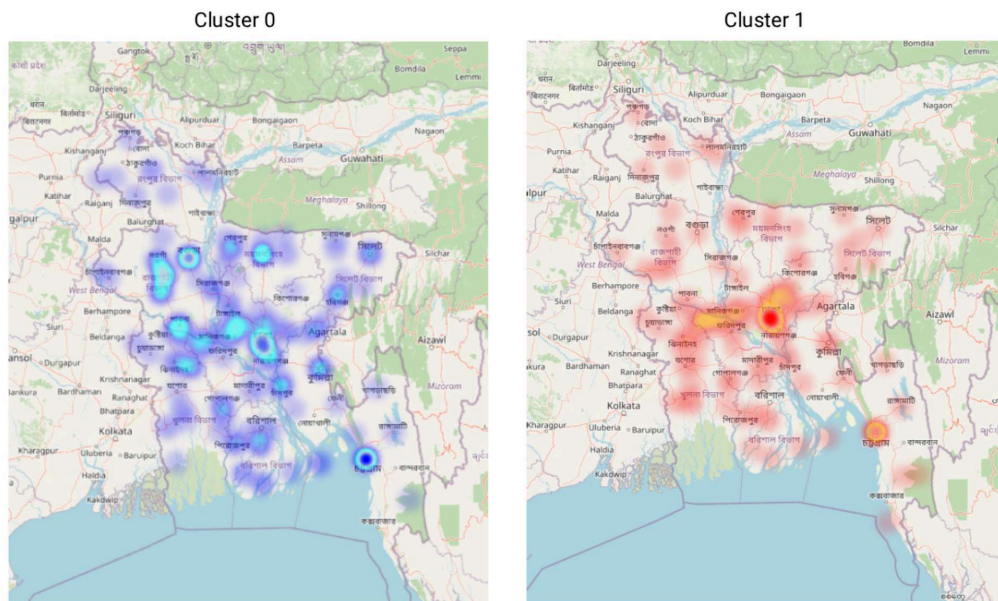


Figure 4.18: Comparing spatial density for two clusters

4.6.5. Temporal Validation and Pattern Discovery

A striking temporal pattern emerged during manual inspection of the cluster assignments. Cluster 0 contains exclusively records from 2020, whereas Cluster 1 consists entirely of cases from 2017–2019 and 2021–2025 (i.e., the years 2017–2025 excluding 2020), as shown in Figure 4.19. This temporal separation aligns precisely with the dataset construction process: the original Kaggle dataset comprised 760 records all drawn from the year 2020, and the 857 manually collected cases covering years before and after 2020 were added later and contained systematically higher missingness across several columns.

Importantly, temporal information was not used as an explicit clustering constraint. The fact that this boundary emerged naturally demonstrates the robustness of the preprocessing, imputation, and clustering pipeline in preserving latent structure. This temporal validation provides strong qualitative evidence that the proposed framework can uncover meaningful, time-specific patterns in complex, partially incomplete data.

4.7. Discussion of Findings

The findings validate the effectiveness of the proposed comparison framework and composite scoring strategy. The balance metric played a critical role in penalising noisy or

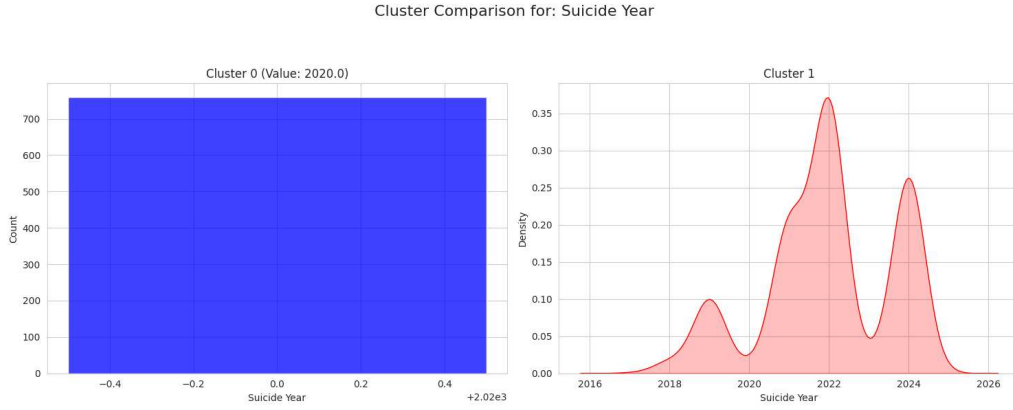


Figure 4.19: Comparing years for two clusters

highly imbalanced solutions, particularly for DBSCAN-based pipelines that achieved deceptively high geometric scores. This prevented misleading interpretations and ensured that only practically meaningful clusterings ranked highly.

Imputation strategy had a substantial impact on performance. MissForest consistently produced the strongest results, both in terms of peak and average composite scores, while pipelines without imputation performed markedly worse. No-context configurations slightly outperformed context-aware ones on average, suggesting that leveraging a richer predictor space during imputation can enhance clustering stability.

The top-ranked pipeline yielded clusters that were coherent across multiple dimensions, including demographics, environment, social context, geography, and time. The emergence of a clean temporal separation without explicit supervision is particularly noteworthy and underscores the practical utility of the framework. Nevertheless, limitations remain, including sensitivity to hyperparameter choices, computational cost, and the absence of external ground-truth labels, which should be addressed in future work.